



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CS&IT
Course Outcome	CO-3

Case Study Topic: Content Recommendation Network Analysis using BFS and DFS in MediaVault

Work Done Summary:

MediaVault recommends related content based on user viewing patterns. Efficient traversal of content relationships is required to identify recommended content. BFS and DFS algorithms were implemented to traverse the recommendation network efficiently.

Problem Statement

The MediaVault system maintains relationships between different content items. Efficient traversal is required to generate recommendations and discover related content.

The challenge is to:

- Traverse content networks efficiently.
- Discover related content.
- Improve recommendation accuracy.
- Support content analytics.
- .

Proposed Solution

Graph traversal algorithms BFS and DFS are used to analyze content relationships.

Implementation Details:

Algorithms Used

- Breadth First Search (BFS)
- Depth First Search (DFS)

Operations Performed

1. Create content relationship graph.
2. Traverse graph using BFS.
3. Traverse graph using DFS.
4. Display traversal order.
5. Analyze recommendation network.

```

public class ContentTraversal {
    static void BFS(String start) {
    }

    static void DFS(String node,
                    Set<String> visited) {
        visited.add(node);

        System.out.print(node);

        for(String neighbour :
            graph.get(node)) {
            if(!visited.contains(neighbour)) {
                System.out.print(" -> ");
                DFS(neighbour, visited);
            }
        }
    }

    public static void main(String[] args) {
        graph.put("Movie_A",
            Arrays.asList(
                "WebSeries_B",
                "Podcast_C",
                "Documentary_D"));
    }
}

```

Result: Screenshots/Output

```

===== MediaVault Recommendation Network =====

BFS Traversal:
Movie_AWebSeries_B -> Podcast_C -> Documentary_DShortFilm_E -> MusicVideo_F

DFS Traversal:
Movie_A -> WebSeries_B -> Podcast_C -> Documentary_D -> ShortFilm_E -> MusicVideo_F

Traversal Completed

```

- BFS successfully traversed content level-wise.
- DFS successfully explored content relationships.
- Recommendation discovery became efficient.
- Content network analysis improved.

GitHub Link:

<https://github.com/snehiths399-byte/MediaVault--Digital-content-streaming-and-Analytics-system>

Student Signature	Faculty Signature